

Blue carbon site screen

Laguna de Términos, Campeche, Mexico · Prepared 2026-10-09 · Habitat model: Unet (resnet34)

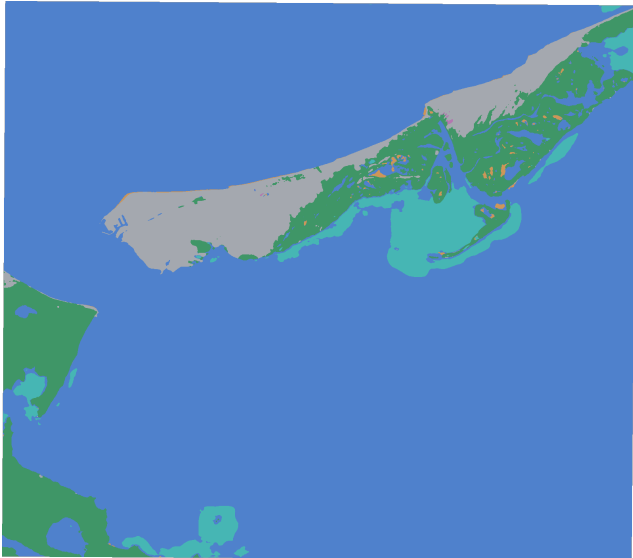
Screening verdict: **Needs investigation**

Why: Inside the Laguna de Términos Flora and Fauna Protection Area, a federal protected area decreed in 1994 (about 706,000 ha).

Satellite image and habitat map



Satellite image (Sentinel-2)



Habitat map

■ Open water ■ Mangrove ■ Salt marsh ■ Seagrass ■ Tidal flat / bare ■ Other land ■ Freshwater wetland

Habitats found

Habitat	Area (ha)	Likely range (ha)
Mangrove	7,682	7,414 – 7,950
Salt marsh	34	31 – 37
Seagrass	5,947	3,232 – 8,662

Carbon

Habitat	Stored (t CO2e)	Absorbed per year (t CO2e)	Soil carbon source
Mangrove	14,692,540 (13,841,321–15,605,522)	46,253	IPCC global default
Salt marsh	33,781 (32,051–36,176)	111	IPCC global default
Seagrass	2,432,292 (2,038,508–2,851,296)	9,668	IPCC global default

Credit potential (illustrative)

Basis	Low	Mid	High
Sequestration of existing habitat (credits per year)	28,727	38,578	51,435
Over 30 years (best basis)	861,822	1,157,342	1,543,050
Value per year at \$20 / \$36.5 / \$50 per credit	\$574,548	\$1,408,099	\$2,571,751

Deductions applied: risk buffer 15%, leakage 10%, uncertainty 10% (total 31%). Credits = area x carbon rate x (1 - deductions), as in the plan; real deductions come from the registry's tools and the methodology.

Change over time

Change study not yet available for this site.

Rights and status screen

Check	Status	Finding
Protected area	Risk	Inside the Laguna de Términos Flora and Fauna Protection Area, a federal protected area decreed in 1994 (about 706,000 ha). Conservation credits must show the project adds protection beyond what the decree already requires (additionality). [https://en.wikipedia.org/wiki/Laguna_de_T%C3%A9rminos]
Ramsar	Caution	Ramsar Wetland of International Importance no. 1356 (designated 2 February 2004, 705,016 ha). Listed threats: sedimentation, pollution, deforestation, declining fish stocks and oil pipelines. [https://rsis Ramsar.org/ris/1356]
Mangrove law	Caution	Mexican law prohibits removing, filling or damaging mangroves (General Wildlife Law, Article 60 Ter). Legal protection already exists, so avoided-loss credits need evidence of real ongoing loss despite it. [https://www.diputados.gob.mx/LeyesBiblio/pdf/LGVS.pdf]
Tenure	Caution	Coastal land is largely federal (federal maritime-terrestrial zone) or communal (ejido). Carbon rights would come from agreements with communities and authorities, not a land purchase. Confirm with the National Agrarian Registry (RAN) and a local lawyer.
Existing projects	OK	No registered blue carbon project for Laguna de Términos was found in a web search on 2026-10-09 (one Verra-listed mangrove project exists in Nayarit). Confirm on the Verra registry before any work. [https://verra.org/projects_public_comm/aztlan-blue-carbon-mangrove-conservation-project-conserving-the-marismas-nacionales-of-nayarit]
Threats	Caution	Pemex operates oil infrastructure inside the protected area; spills and pipeline construction are recurring threats. This is evidence of pressure (relevant to additionality) and a permanence risk. [https://en.wikipedia.org/wiki/Laguna_de_T%C3%A9rminos]

Methods and limits

- Habitats are mapped from free Sentinel-2 imagery with a deep-learning model (BlueCarbon-AI), trained on 39 coastal sites and scored on areas it never saw.
- Map accuracy on unseen areas for this habitat: F1 0.97. Areas are corrected for the model's known errors; the range shown is a plain-language margin, not a statistical interval.
- Carbon: soil carbon from measured cores where available (Smithsonian Coastal Carbon Library), otherwise IPCC 2013 Wetlands Supplement Tier 1 defaults; biomass and burial rates IPCC Tier 1.
- Seagrass is mapped conservatively and is a lower bound; murky or deep meadows are often missed.

Screening and estimates only. Not legal, financial or investment advice. Only a registry issues credits, after independent validation and verification. Confirm land tenure and carbon rights with a local lawyer.